

1	Course Name:
	Mobile and Ubiquitous Computing
2	Course Code:
	ISB26603
3	Year and Semester Offered:
	Year 2 Semester 2
4	Credit Hours Value:
	3 Credit Hours [3 SLT Credit]
5	Prerequisite(s), if applicable:
	ISB16003 Object Oriented Programming
6	Course Synopsis:
	This course aimed at providing students to understand the state-of-the-art of mobile and ubiquitous computing technologies, and to get familiar with the development of application software for mobile applications. Mobile and ubiquitous computing will be discussed from three perspectives: mobile technology, application development, and user interaction. Students will work on practical projects that address cutting-edge real-world problems and will develop innovative solutions to them through means of mobile and ubiquitous computing.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Identify and solve problems based on the main theory of mobile and ubiquitous computing. 2. Analyse the capabilities of different component mobile and ubiquitous computing technology. 3. Build mobile application that integrates design interfaces and features functionalities for intended users.
8	Content Outline:
	Topic
	1. Ubiquitous Computing Concepts, Architecture and Applications
	2. Mobile Application Environment
	3. Mobile Application Design - Human Computer Interaction
	4. Mobile Application - Graphics and Multimedia
	5. Mobile Application - Context Aware and Smart Environment
	6. Mobile Application - Data Management
	7. Mobile Application - Security Issues
	8. Lab 1: Setting mobile application development environment
	9. Lab 2: Introduction to mobile application development
	10. Lab 3: Mobile Application- UI and UX
	11. Lab 4: Mobile Application - Multimedia, graphic and Animation
	12. Lab 5: Mobile Application - Application Program Interface (API)

	13. Lab 6 : Mobile Application - Database
	14. Mini project Preparation
9	References:
	1. Raywenderlich Tutorial Team, Flutter Apprentice (Second Edition): Learn to Build Cross-Platform Apps, Razeware LLC, 2021 [ISBN-13: 978-1950325481] 2. Lawrence T.Yang, Evi Syukur, Seng W. Loke. Handbook on Mobile and Ubiquitous Computing. Taylor & Francis, CRC Press. 2016. [eBook ISBN 9781439848128]